ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C2 · SOUND

What vibrates?

★☆☆
Grade 3

Learning goals

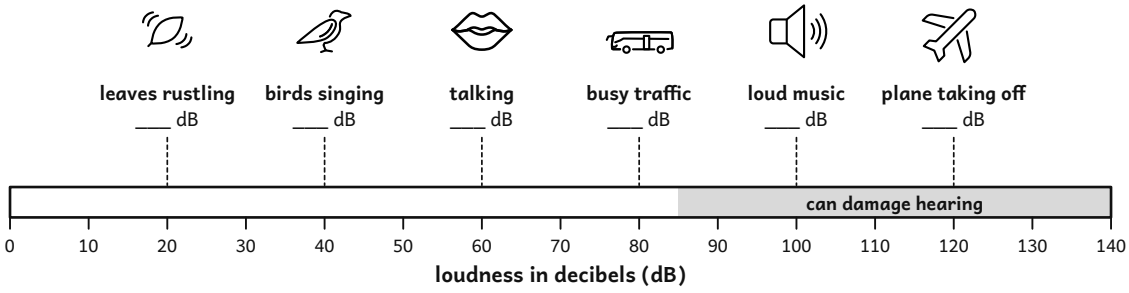
- Name what vibrates.
- Read the decibel scale.

1 Draw a line to match the instrument to the part that vibrates. [4]

Draw a line to match each pair, or write the letter.

dhol ●	→ A	● **A** the drum skin
ektara ●	→ C	● **B** the air inside
bansuri ●	→ B	● **C** the string
your voice ●	→ D	● **D** the voice box in your throat

2 Write the loudness of each sound. [3]



leaves	birds	talking	traffic	loud music	plane
20 dB	40 dB	60 dB	80 dB	100 dB	120 dB

3 True or false? [3]

Statement	True	False	If false, correct it
1. Sound can travel through water.	✔		
2. Sound travels through space.		✔	There is no air in space to carry it.
3. A loud sound always has a high pitch.		✔	Loudness and pitch are different.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C2 · SOUND

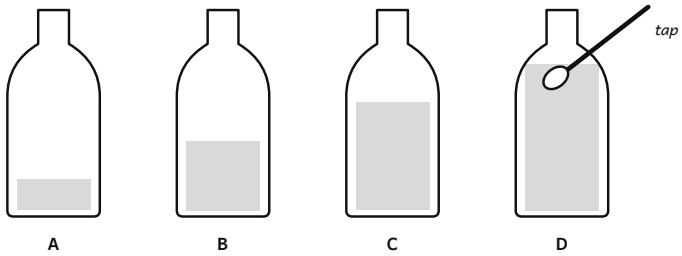
Thinking about sound

★★☆
Grade 3

Learning goals

- Explain pitch and loudness.

1 **Which bottle makes the highest note when tapped? Which makes the lowest?** [2]



Highest: A (least water). Lowest: D (most water).

2 **An ektara player presses the string to make it shorter. What happens to the pitch?** [1]

It gets higher.

3 **Why do people who work with loud machines wear ear defenders?** [2]

Loud sounds above 85 dB can damage hearing; ear defenders block the sound.

4 **A string phone only works when the string is tight. Why?** [2]

A tight string carries the vibrations; a loose string does not vibrate well.

ANSWER KEY

INVESTIGATE · C2 · SOUND

Which material blocks sound best?

★★☆ Grade 3

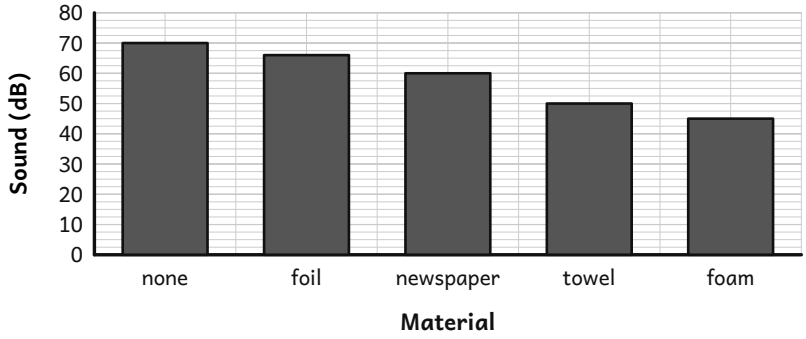
Learning goals

- Measure sound in decibels.
- Draw a bar chart and conclude.

Class 3C put a ringing alarm clock inside a box and wrapped it in different materials. They measured the sound 1 metre away with a sound meter app.

1 **Draw a bar chart.** [3]

Material	none	foil	newspaper	towel	foam
Sound (dB)	70	66	60	50	45



2 **Which material blocked the sound best?** [1]

foam

3 **What kind of materials are good sound insulators?** [2]

soft, thick materials like foam and towels

4 **What did they keep the same to make it fair?** [1]

same alarm clock, same box, same distance (1 metre)

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C2 · SOUND

Too much noise!

★★★
Grade 3

Learning goals

- Use what you know about sound to solve a real problem.

Science | Health | Citizenship

Dhaka is one of the noisiest cities in the world. On busy roads the sound is often above **85 dB** because of car horns, buses, building work and loudspeakers. Traffic police stand in this noise all day.

Too much noise can damage hearing and make it hard to sleep, learn and think. Bangladesh has rules: very loud hydraulic horns are not allowed, and roads near **schools and hospitals** are **silent zones** where drivers must not honk.

Based on reports about noise pollution in Dhaka and Bangladesh's noise rules.

1 **Name three things that make Dhaka noisy.** [3]

car horns, buses, building work, loudspeakers

2 **Why are roads near hospitals and schools silent zones?** [2]

Sick people need rest; children need quiet to learn.

3 **Make a "Do not honk" sign for your school gate.** [2]

Silent zone sign

e.g. a horn with a line through it: "Silent zone: no horns".

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C2 · SOUND

Show what you know

★★★
Grade 3

Learning goals

- Use sound words.

1 Find the words.

J	U	L	O	U	D	V	Y	P
E	S	L	W	V	T	V	G	I
H	A	U	Y	I	W	K	D	T
D	I	R	U	B	T	G	E	C
N	R	L	G	R	I	N	C	H
O	S	U	R	A	M	B	I	I
I	V	O	M	T	H	W	B	U
S	K	G	W	E	B	B	E	K
E	N	J	B	V	D	O	L	H

decibel drum ear loud noise pitch vibrate

2 Loudness is measured in... [1]

○ metres
● **decibels**
○ grams

3 A short, tight string makes a... [1]

● **high pitch**
○ low pitch
○ no sound

4 Challenge. Design a quiet corner for your classroom. Which materials would you use? [2]

curtains, carpet, cushions, foam on the walls: soft, thick materials soak up sound.